

Growth inhibition and induction of apoptosis in NB4 promyelocytic leukemia cells by trypsin inhibitor from sweet potato storage roots.

The objective of this study was to investigate the antiproliferative effect and the mechanism of trypsin inhibitor (TI) from sweet potato [*Ipomoea batatas* (L.) Lam. 'Tainong 57'] storage roots on NB4 promyelocytic leukemia cells. The results showed that TI inhibited cellular growth of NB4 promyelocytic leukemia cells in a time-dependent and dose-dependent manner, and treatment for 72 h induced a marked inhibition of cellular growth, showing an IC₅₀ of 57.1 ± 8.26 µg/mL. TI caused cell cycle arrest at the G1 phase as determined by flow cytometric analysis and apoptosis as shown by DNA laddering. TI-induced cell apoptosis involved p53, Bcl-2, Bax, and cytochrome c protein in NB4 cells. P53 and Bax proteins were accumulated, and antiapoptotic molecule Bcl-2 was decreased in the tested cells in a time-dependent manner during TI treatment. TI also induced a substantial release of cytochrome c from the mitochondria into the cytosol. Hence, TI induced apoptosis in NB4 cells through a mitochondria-dependent pathway, which was associated with the activation of caspase-3 and -8. These results demonstrated that TI induces NB4 cell apoptosis through the inhibition of cell growth and the activation of the pathway of caspase-3 and -8 cascades.